

ONLINE BULK OPTIMIZATION SYSTEM
WEBSITE AI23431- WEBTECHNOLOGY
AND MOBILE APPLICATION
MINI-PROJECT REPORT

SUBMITTED BY

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in partial fulfillment for the award of the degree of

BACHELOR OF TECHNOLOGY

in

ARTIFICIAL INTELLIGENCE AND DATA SCIENCE



**DEPARTMENT OF ARTIFICIAL INTELLIGENCE
AND DATA SCIENCE**

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RAJALAKSHMI ENGINEERING COLLEGE

(an Autonomous Institution Affiliated to Anna University, Chennai)

BONAFIDE CERTIFICATE

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Humeshkumar K E- 241801095

DEPARTMENT VISION

To promote highly Ethical and Innovative Computer Professionals through excellence in teaching, training and research.

DEPARTMENT MISSION

- To produce globally competent professionals who are motivated to learn emerging technologies and demonstrate innovation in solving real-world problems.
- To promote research activities among students and faculty members that contribute meaningfully to society.
- To instill strong moral and ethical values in professional practice.

PROGRAMME EDUCATIONAL OBJECTIVES(PEO'S)

PEO 1: To equip students with an essential background in computer science, basic electronics and applied mathematics.

PEO 2: To prepare students with fundamental knowledge in programming languages, and tools and enable them to develop applications.

PEO 3: To encourage the research abilities and innovative project development in the field of AI, ML, DL, networking, security, web development, Data Science and also emerging technologies for the cause of social benefit.

PEO 4: To develop professionally ethical individuals enhanced with analytical skills, communication skills and organizing ability to meet industry.

PROGRAMME OUTCOMES (POs)

PO 1: Engineering knowledge: Apply the knowledge of Mathematics, Science, Engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.

PO 2: Problem analysis: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.

PO 3: Design/development of solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.

PO 4: Conduct investigations of complex problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

PO 5: Modern tool usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.

PO 6: The engineer and society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.

PO 7: Environment and sustainability: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.

PO 8: Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.

PO 9: Individual and team work: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

PO 10: Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

PO 11: Project management and finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

PO 12: Life-long learning: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

PROGRAM SPECIFIC OUTCOMES (PSOs)

A graduate of the Artificial Intelligence and Data Science Program will demonstrate

PSO 1: Foundation Skills: Ability to understand, analyze and develop computer programs in the areas related to algorithms, system software, web design, AI, machine learning, deep learning, data science, and networking

PSO 2: Problem-Solving Skills: Ability to apply mathematical methodologies to solve computational task, model real world problem using appropriate AI and ML algorithms. To understand the standard practices and strategies in project development, using open- ended programming environments to deliver a quality product.

PSO 3: Successful Progression: Ability to apply knowledge in various domains to identify research gaps and to provide solution to new ideas, inculcate passion towards higher studies, creating innovative career paths to be an entrepreneur and evolve as an ethically social responsible AI and ML professional.

COURSE OBJECTIVE

- To provide foundational knowledge and practical skills in creating and structuring web pages using HTML, enabling students to build accessible and well-organized websites.
- To understand and practice Embedded Dynamic Scripting on Client-side Internet Programming.
- To implement Server Side Scripting.
- To facilitate students to understand android Application Design
- To help students to gain a basic understanding of Android application development

COURSE OUTCOME

On completion of course students will be able to,

- CO 2: Upon completing the course, students will be able to design and develop basic web pages with proper structure and semantic elements, ensuring accessibility and functionality.
- CO 3: Design and implement dynamic web page with validation using javascript objects and by applying different event handling mechanism.
- CO 4: Design and implement simple webpage to learn JSP and Servlet.
- CO 5: Design and implement simple Application Design.
- CO 6: Identify various concepts of mobile programming that make it unique from programming for other

CO-PO-PSO Mapping

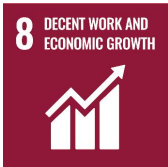
CO	PO 1	P O 2	P O 3	P O 4	P O 5	P O 6	P O 7	P O 8	P O 9	P O 10	P O 11	P O 12	PS O 1	PS O 2	PS O 3
CO 1	3	3	3	3	3	2	2	2	3	2	3	3	3	3	3
CO 2	3	3	3	3	3	2	-	-	2	2	2	3	3	2	2
CO 3	3	3	3	2	3	1	1	2	3	3	3	3	3	3	3
CO 4	3	3	3	3	3	2	1	2	3	2	2	3	3	3	3
CO 5	1	1	1	1	1	-	-	-	3	3	3	3	1	-	2

Note: Correlation levels 1, 2 or 3 are as defined below:

1: Slight (Low) 2: Moderate (Medium) 3:

Substantial (High) No correlation: “-”

SUSTAINABLE DEVELOPMENT GOALS

SDG	GOAL	JUSTIFICATION
 SDG-4	Quality Education	Education drives progress on other SDGs, breaks poverty cycles, and promotes peace and inequality reduction.
 SDG-8	Decent Work and Economic Growth	Necessary for social stability, shared prosperity, and combating poverty.
 SDG-9	Industry, Innovation, and Infrastructure	Infrastructure is vital for economic growth and stability. It acts as a poverty reduction tool through job creation and provides technological solutions to environmental challenges.
 SDG-17	Partnerships for the Goals	Given the interconnected nature of the SDGs, success requires collaborative action, global resource sharing, and bridging the immense financing gap faced by developing nations.

ABSTRACT

This project is an online bulk optimization website designed to simplify the process of searching, adding, and inquiring about real estate. It provides a convenient platform where users can explore available properties, filter them based on location, and connect with property owners.

The application allows users to create accounts, log in, and manage their own listings. Users can add new properties with details such as price, location, description, and images, as well as view complete information about each listing. A contact feature is also included, enabling users to send inquiries directly to property owners.

The system is developed using Java, JSP, and MySQL, and follows a structured approach to keep different parts of the application organized. Basic security features such as user authentication and safe database operations are implemented to ensure reliability.

Overall, the project demonstrates how web technologies can be used to build a practical and user-friendly platform for real estate browsing and management.

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
HTML	HyperText Markup Language
CSS	Cascading Style Sheets
JSP	Java Server Pages
JDBC	Java Database Connectivity
SQL	Structured Query Language
DBMS	Database Management System
UI	User Interface
UX	User Experience
HTTP	HyperText Transfer Protocol

CHAPTER 1

INTRODUCTION

1.1 General

In today's digital era, the internet has become a primary medium for accessing information and services. The real estate sector is no exception, as more people prefer to search for properties online rather than relying on traditional methods such as brokers or newspaper advertisements. Online platforms provide users with quick access to a wide range of property listings, making the process of buying, selling, or renting properties more efficient and convenient.

This project focuses on the development of an online bulk optimization system that allows users to monitor available inventory, view detailed information, and manage stock efficiently. The system is designed to be user-friendly and accessible, enabling users to perform various actions such as searching for products based on category, adding new stock, and sending alerts for low inventory.

1.2 Need for Online Bulk Optimization System

Traditional grocery inventory management methods are often manual, time-consuming, and difficult to maintain efficiently. Grocery store owners frequently face issues such as inaccurate stock tracking, product wastage, delayed restocking, and inefficient bulk inventory handling. These problems can lead to financial losses and reduced customer satisfaction.

An online bulk optimization system addresses these challenges by providing a centralized platform where multiple users can interact. It reduces dependency on manual tracking, improves accuracy, and allows users to manage different inventory items easily.

Additionally, it ensures that information is updated in real time, making the process more reliable and efficient.

1.3 Scope of the System

The Grocery Bulk Optimization System is designed to provide an efficient and user-friendly platform for managing grocery inventory and bulk stock operations. The system mainly focuses on helping grocery store owners and administrators monitor products, manage stock levels, and improve inventory handling through automation and analytics.

- Monitor products.
- Manage stock levels.
- Improve inventory handling through automation and analytics

1.4 Organization of the Report

This report is organized into several chapters that describe the development and implementation of the online property listing system.

Chapter 1: provides the introduction to the project, including the general overview, need for the system, and scope of the project.

Chapter 2: presents the literature survey, which discusses existing real estate platforms and related work.

Chapter 3: explains the problem formulation and objectives of the proposed system.

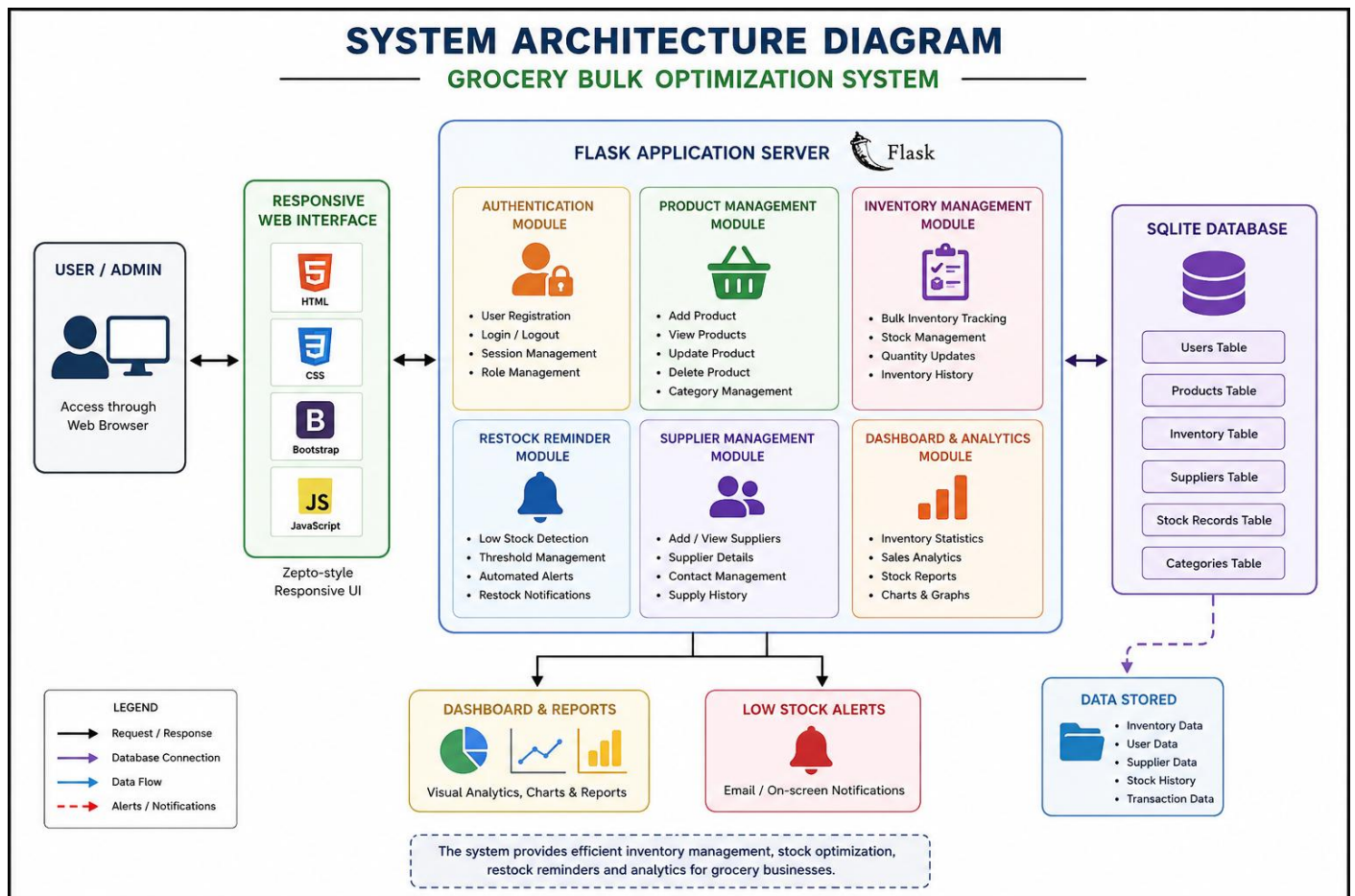
Chapter 4: describes the system design, including system architecture, database design, and system requirements.

Chapter 5: discusses the implementation of the system and explains the different modules.

Chapter 6: presents the results and discussion, including system outputs and performance evaluation.

Chapter 7: concludes the project and suggests possible future enhancements.

System architecture diagram:



CHAPTER 2

LITERATURE SURVEY

2.1 Introduction

The literature survey explores existing systems and approaches used in online property listing and real estate platforms. With the rapid growth of internet usage, digital solutions have become the preferred method for property search and management. This section examines how current platforms function, the technologies they use, and the challenges they face. Understanding these aspects helps in designing a system that is both efficient and user-friendly while addressing the limitations of existing solutions.

2.2 Existing Bulk Optimization Systems

Existing bulk optimization systems, particularly in the context of inventory management for businesses like grocery stores, often rely on traditional methods that present significant challenges. These systems typically involve manual tracking, which is prone to human error and can be time-consuming.

The limitations of these traditional bulk optimization systems include inaccurate stock tracking, leading to product wastage and financial losses. Delayed restocking due to inefficient processes can also result in lost sales and reduced customer satisfaction. Furthermore, the lack of real-time data and centralized platforms makes it difficult for businesses to efficiently handle large inventories and adapt to changing demands.

2.3 Related Work

Previous studies and projects in the domain of web-based property management systems have focused on improving usability, efficiency, and accessibility. Researchers have explored the use of dynamic web technologies to create interactive user interfaces and faster data retrieval

systems. Database-driven applications are commonly used to manage large volumes of property data, ensuring that users can access updated and accurate information.

Security and user authentication are also key areas of focus in related work, with systems implementing login mechanisms and session management to protect user data. Some solutions incorporate advanced features such as recommendation systems and location tracking to enhance user experience. However, these features often increase system complexity and may not be necessary for all users, especially in smaller-scale applications.

2.4 Summary

The analysis of existing bulk optimization systems and related work highlights the importance of providing a balanced system that combines robust functionality with operational simplicity. While many advanced systems offer a wide range of features, there remains a critical need for applications that are easy to use, efficient, and precisely focused on essential inventory management functionalities.

Based on this study, the proposed system aims to deliver a streamlined bulk optimization platform that enables users to monitor products, manage stock levels, and improve inventory handling with ease. The focus is on creating a reliable and user-friendly application that meets core requirements for efficient inventory management without unnecessary complexity.

CHAPTER 3

PROBLEM FORMULATION AND OBJECTIVES

3.1 Problem Statement

In traditional inventory management for businesses like grocery stores, bulk stock tracking and optimization are often carried out through manual methods or outdated systems. These approaches are frequently time-consuming, prone to human error, and lack real-time transparency. Store owners and managers may struggle to find accurate and up-to-date information on stock levels, leading to issues such as product wastage, missed sales opportunities due to stockouts, and inefficient ordering processes.

Existing inventory management platforms, while some offer advanced features, are often complex and may include unnecessary functionalities for basic users, making them difficult to implement and navigate. They can also be costly, requiring significant investment in hardware and software, and may not provide a simple, centralized interface for managing diverse inventory items and supplier interactions. Additionally, many systems lack the ability to provide real-time alerts for low stock or expiring products, further hindering efficient bulk optimization.

Therefore, there is a critical need for a streamlined and user-friendly web-based system that allows users to easily monitor, search, add, and manage bulk inventory, while providing real-time data and alerts to optimize stock levels and reduce operational inefficiencies.

3.2 Objectives of the System

The main objective of the system is to develop a web-based application that simplifies the process of bulk inventory management and optimization. The specific objectives are as follows:

- To design and develop a user-friendly interface for monitoring and managing bulk inventory
- To enable users to search for products based on category or other criteria
- To provide functionality for users to add and manage their own stock listings
- To allow users to view detailed information about each product
- To implement a feature for sending alerts for low stock or expiring products
- To develop a secure login and registration system using session management

CHAPTER 4

SYSTEM DESIGN

4.1 Introduction

System design is an important phase in software development where the overall structure and components of the system are planned. It defines how different modules interact with each other and how data flows within the system. A well-designed system improves efficiency, reliability, and ease of maintenance.

The online property listing system is designed as a web-based application that allows users to browse, search, and manage property listings. It provides features such as user registration, login, property posting, and inquiry handling. The system follows a client-server architecture where users interact through web pages, while the backend processes requests and communicates with the database.

4.2 System Architecture

The system follows a three-layer architecture consisting of the presentation layer, application layer, and data layer.

Presentation Layer

This layer represents the user interface where users interact with the system. It includes web pages such as the homepage, login page, registration page, property listing page, and search page. These pages are developed using JSP, HTML, and CSS, and are responsible for displaying data to users and collecting input.

Application Layer

This layer handles the core logic of the system. It processes user requests, manages sessions, performs validation, and controls the flow of data between the presentation layer and the database.

The backend is implemented using Flask, where each route is responsible for a specific functionality:

- `/products` → Displays all inventory products
- `/add_product` → Adds new inventory items
- `/product_detail/` → Shows detailed product information
- `/alerts` → Manages low inventory alerts
- `/login&/register` → Handle user authentication
- `/delete_product/` → Deletes inventory products

This layer ensures proper execution of all operations and maintains system functionality.

Data Layer

This layer is responsible for storing and managing all application data. The system uses a MySQL database connected through JDBC for data operations.

The database consists of three main tables:

- **users** → Stores user details such as name, email, and password
- **products** → Stores property information including title, price, location, description, image, and user ID
- **inquiries** → Stores user inquiries related to properties

4.3 Database Design

The database is designed to efficiently store and manage data related to users, properties, and inquiries. Each table is structured with appropriate fields and relationships.

- The **users** table stores registered user information
- The **products** table stores all property listings and is linked to users through a foreign key
- The **inquiries** table stores messages sent by users regarding specific properties

Relationships between tables ensure data consistency and integrity. For example, each property is associated with a user, and each inquiry is linked to a specific property.

4.4 System Requirements

System requirements define the hardware and software resources needed to run the application successfully.

4.4.1 Hardware Requirements

The system does not require high-end hardware and can run on standard computers.

- **Processor:** Intel Core i3 or higher
- **RAM:** Minimum 4 GB
- **Storage:** 500 GB Hard Disk or SSD
- **Input Devices:** Keyboard and Mouse
- **Internet Connection:** Required

4.4.2 Software Requirements

The system is developed using commonly available tools and technologies.

- 5 **Operating System:** Windows 10 or later
- 6 **Frontend Technologies:** HTML, CSS, JSP
- 7 **Backend Technology:** Java Servlets
- 8 **Database:** MySQL
- 9 **Database Connectivity:** JDBC
- 10 **Web Server:** Apache Tomcat 9.0
- 11 **Development Environment:** Eclipse IDE for Enterprise Java
- 12 **Web Browser:** Google Chrome or any modern browser

CHAPTER 5

SYSTEM IMPLEMENTATION

5.1 System Methodology

The system is developed using a structured approach based on the Model-View-Controller (MVC) architecture. This methodology helps in separating the user interface, business logic, and data handling, making the system easier to develop and maintain.

- **Model:** Represents the data and business logic using Java classes such as [Property](#), [User](#), and [Inquiry](#).
- **View:** Consists of JSP pages that display data to users and collect input.
- **Controller:** Implemented using Java Servlets/FLask, which handle user requests, process data, and control the flow between the view and model.

The development process involves designing the database, creating model classes, implementing servlets for different functionalities, and developing JSP pages for user interaction. JDBC is used to connect the application with the MySQL database.

5.2 Module Descriptions

The system is divided into several modules, each responsible for a specific functionality.

5.2.1 User Module

This module handles user registration, login, and session management.

- New users can register by providing details such as name, email, and password
- Existing users can log in using valid credentials
- Session management is implemented using [HttpSession](#)
- User data is stored securely in the database

5.2.2 Product Management Module

This module allows users to add, view, and delete product listings.

- Users can add new products with details such as name, category, quantity, description, and image.
- Product listings are stored in the database.
- Users can delete their own product listings.
- All products are displayed on the homepage.

5.2.3 Product Search Module

This module provides functionality to search for products based on category or name.

- Users can enter a search keyword.
- The system performs a search using SQL LIKE queries.
- Matching results are displayed dynamically.
- Helps users quickly find relevant products.

5.2.4 Product Details Module

This module displays detailed information about a selected product.

- Shows complete details such as name, category, quantity, description, and image.
- Allows users to view full product information before making decisions.
- Improves user experience by providing clarity.

5.2.5 Alerts Module

This module enables users to receive notifications for low stock or expiring products.

- Users can set custom alert thresholds for stock levels.
- Alert details are stored in the database.
- Helps prevent stockouts and minimizes product wastage.

5.2.6 Login and Authentication Module

This module ensures secure access to the system.

- Validates user credentials during login.
- Uses session management to maintain user state.
- Prevents unauthorized access to restricted features.
- Handles login and logout operations.

5.2.7 Delete Product Module

This module allows users to remove their product listings.

- Only the owner of the product can delete it.
- Related alerts are also removed from the database.
- Maintains data consistency and integrity.

CHAPTER 6

RESULTS AND DISCUSSION

6.1 System Output Screens

The developed system was tested by executing all major functionalities, and the outputs were verified through different user interactions. The application successfully displays dynamic content retrieved from the database and responds correctly to user inputs.

The main output screens of the system include:

- **Homepage (Property Listings):**

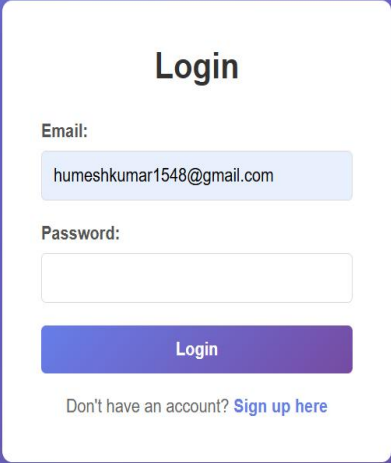
Displays all available property listings with details such as dashboard, addproduct, reports and settings. The data is fetched from the database and shown in a structured format.

BulkOpt					Dashboard	Add Product	Reports	Settings
Bulk Inventory Optimization					Manage stock efficiently and avoid shortages			
Product Inventory								
ID	Product	Quantity	Restock Cycle	Action				
2	ATTA	5	3 days	Delete				
3	INDIA GATE BASMATHI RICE	5	3 days	Delete				
4	SVS KADALAI OIL	4	Monthly	Delete				
5	CYNTHOL SOAP	8	Monthly	Delete				
7	UDHAYAM URAD DHALL	4	Monthly	Delete				
8	UDHAYAM TOOR DHALL	4	Monthly	Delete				
9	UDHAIYAM DHALL MOONG	1	Weekly	Delete				

- **Login Page:**

Allows registered users to log in using their email and password. Proper validation messages are displayed for incorrect credentials.

The login page ensures secure access to the system, protecting user data and restricted functionalities. It also includes options for password recovery and new user registration, enhancing user convenience and accessibility.



Login

Email:
humeshkumar1548@gmail.com

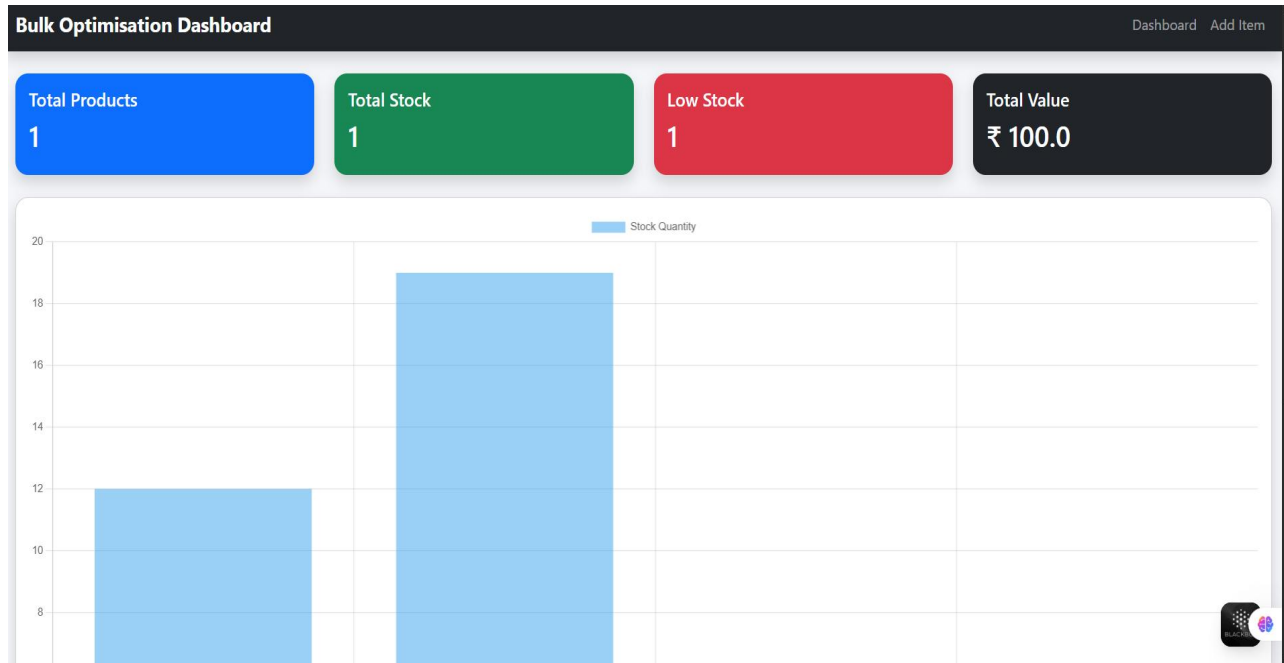
Password:

Login

Don't have an account? [Sign up here](#)

- **Dashboard Page:**

Shows complete information about a selected property, including description and image.



- **Add Product Page:**

Allows users to add new product listings with details and image upload functionality. This includes fields for product name, description, category, unit price, quantity in stock, number of packets per unit (if applicable), and a restock threshold to trigger alerts when inventory is low.

.

Bulk Optimization System - Add Product

Product Name

milK

Quantity

1

l

Packets

3

Restock

3 days

Add Product

These output screens demonstrate that the system is functioning correctly and providing the intended features.

6.2 Performance Evaluation

The system was evaluated based on responsiveness, accuracy, and usability.

- **Response Time:**

The system responds quickly to user requests such as loading property listings, searching, and displaying details.

- **Data Accuracy:**

All data retrieved from the database is displayed correctly without inconsistencies.

- **Usability:**

The user interface is simple and easy to navigate, allowing users to perform actions without confusion.

- **Efficiency:**

The use of JDBC and optimized SQL queries ensures efficient data handling and retrieval.

Overall, the system performs well for small to medium-scale usage and provides a smooth user experience.

6.3 System Testing

The system was tested using different scenarios to ensure reliability and correctness.

- **Functional Testing:**

All features such as registration, login, product addition, search, and inquiry were tested and found to be working as expected.

- **Validation Testing:**

Input fields were tested for correct validation, ensuring that invalid or empty data is not accepted.

- **Authentication Testing:**

Login and session management were tested to ensure secure access to the system.

- **Database Testing:**

Data insertion, retrieval, and deletion operations were verified to ensure proper interaction with the database.

- **Error Handling:**

The system handles errors gracefully and displays appropriate messages to the user.

The testing results confirm that the system is stable, reliable, and ready for deployment.

CHAPTER 7

CONCLUSION AND FUTURE WORK

7.1 Conclusion

The online bulk optimization system was successfully designed and developed to simplify the process of searching, adding, and managing inventory management. The application provides a user-friendly interface that allows users to browse products, search based on location, view detailed information, and communicate with suppliers through inquiries.

The system effectively integrates frontend and backend technologies using Java Servlets, JSP, and MySQL. It follows a structured architecture that ensures clear separation of components, making the application easy to understand and maintain. Key features such as user authentication, product management, and database interaction have been implemented successfully.

The project demonstrates the practical application of web development concepts and provides a reliable platform for handling real estate listings. Overall, the system meets its objectives and performs efficiently for small to medium-scale usage.

7.2 Future Enhancements

Although the system meets current requirements, it can be further improved by adding more advanced features in the future:

- **Advanced Search Filters:**
Adding filters such as price range, and amenities for more precise results.
- **Online Booking or Payment System:**
Allowing users to book or reserve properties directly through the platform.
- **Mobile Application:**
Developing a mobile version of the system for improved accessibility.
- **Admin Panel:**
Providing an admin interface to manage users, properties, and inquiries.
- **Email Notifications:**
Sending automatic notifications for inquiries and updates.
- **Improved Security:**
Enhancing security with encryption and stronger authentication mechanisms.

These enhancements can make the system more scalable, interactive, and suitable for real-world deployment on a larger scale.

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APPENDICES

Appendix-1: Sample Database Tables

Users Table

Field Name	Data Type	Description
user_id	INT (PK)	Unique user ID
name	VARCHAR	User name
email	VARCHAR	User email
password	VARCHAR	User password

Products Table

Field Name	Data Type	Description
product_id	INT (PK)	Unique product ID
title	VARCHAR	Product title
quantity	DOUBLE	Product quantity
packets	VARCHAR	Product packets
restock	TEXT	Product description
user_id	INT (FK)	Reference to user

Alerts Table

Field Name	Data Type	Description
inquiry_id	INT (PK)	Unique inquiry ID
property_id	INT (FK)	Related property
restock	INT(FK)	Days remainder
product	TEXT	Product message

Appendix-2: Sample SQL Queries

-- Insert User

```
INSERT INTO users (name, email, password)
VALUES ('John Doe', 'john@example.com', '123456');
```

-- Insert Product

```
INSERT INTO products (title, price, quantity, packets, restock, user_id) VALUES ('Milk', 2.99,
100, 6, 10, 1);
```

-- Search Product

```
SELECT * FROM products WHERE title LIKE '%Milk%';
```

-- Delete Product DELETE FROM

```
products WHERE product_id = 1;
```

Appendix-4: Key Servlet Example

@WebServlet("/index")

```
public class ProductServlet extends HttpServlet {
```

```
    protected void doGet(HttpServletRequest request,
        HttpServletResponse response)
        throws ServletException, IOException {
```

```
        List<Product> list = new ArrayList<>(); Connection con = DBConnection.getConnection();
```

```
        try {
            Statement stmt = con.createStatement();
            ResultSet rs = stmt.executeQuery(
                "SELECT * FROM products");
```

```
            while (rs.next()) {
                Product p = new Product();
                p.setTitle(rs.getString("title"));
                p.setPrice(rs.getDouble("price"));
                list.add(p);
            }
```

```
        } catch (Exception e)
        { e.printStackTrace();
        }
```

```
request.setAttribute("propertyList", list);
request.getRequestDispatcher("index.jsp")
    .forward(request, response);
}
```

PSEUDOCODE

The following pseudocode represents the core functionalities of the online property listing system.

1. User Registration

```
BEGIN
  INPUT name, email, password

  CONNECT to database

  CHECK if email already exists
  IF exists THEN
    DISPLAY "Email already registered"
  ELSE
    INSERT user details into users table
    DISPLAY "Registration successful"
  ENDIF

  CLOSE database connection
END
```

2. User Login

```
BEGIN
  INPUT email, password

  CONNECT to database

  FETCH user where email and password match

  IF user exists THEN
    CREATE session
    STORE user details in session
    REDIRECT to homepage
  ELSE
    DISPLAY "Invalid
credentials"
  ENDIF

  CLOSE database connection
END
```

3. Add Product

```
BEGIN
  CHECK if user is logged in

  INPUT title, price, quantity, packets, restock,
  image
  CONNECT to database

  INSERT product details into products table with
    user_id from session
  DISPLAY "Product added successfully"
  CLOSE database connection
END
```

4. View Product

```
BEGIN
  CONNECT to database

  FETCH all products ordered by latest

  STORE results in list

  SEND list to JSP page

  DISPLAY products to user

  CLOSE database connection

END
```

5. Send alert

```
BEGIN
  INPUT name, message, product_id CONNECT

  to database

  INSERT alert into alerts table DISPLAY
  "Alert sent successfully" CLOSE database
  connection
END
```

6. Delete Product

```
BEGIN
  GET product_id CHECK if user is owner

  CONNECT to database

  DELETE related alerts from alerts table

  DELETE product from products table DISPLAY

  "Product deleted successfully

  CLOSE database connection
END
```




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